



CAPSTONE PROJECT · TRACK C

# How well can we forecast next month's reservoir pressure using production and injection data?

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Machine Learning for Petroleum Engineers & Geoscientists · Final Project Presentation



# Why pressure forecasting matters

## THE QUESTION

Can we forecast reservoir pressure for next month from injection and production history?

*Forecasting pressure is important to monitor pressure declines before they become a problem and force costly interventions*



## Who uses the answer

Reservoir engineers, production engineers



## The decision it supports

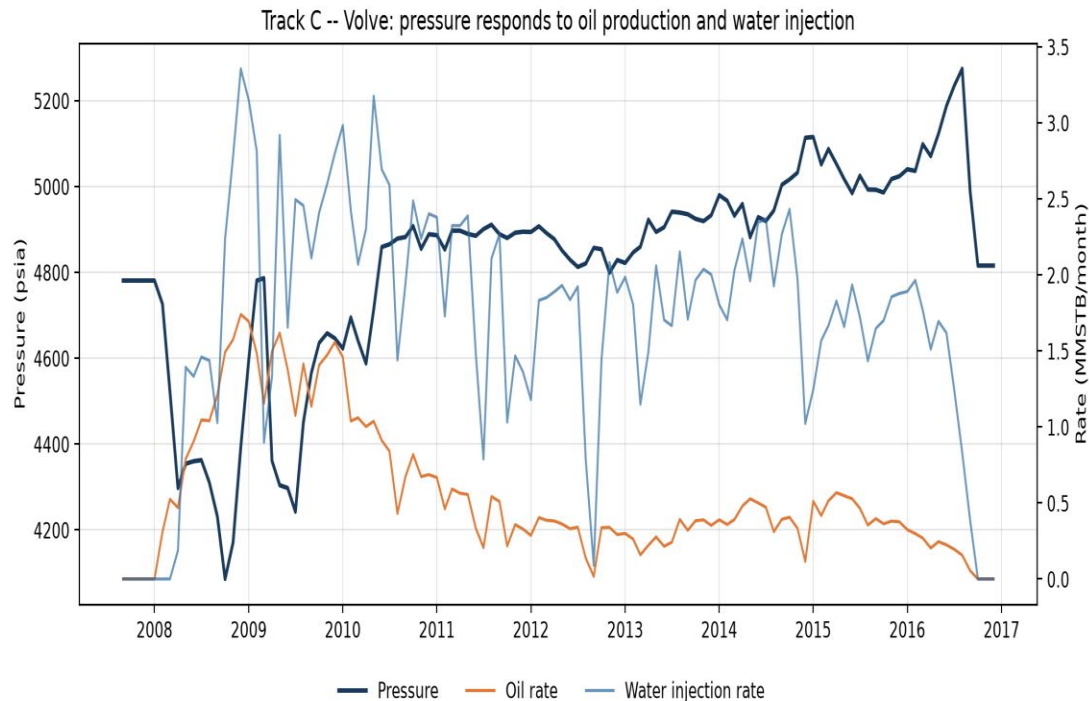
Deciding when to adjust water injection rates, by catching pressure declines early



## Success criterion

A model that beats the persistence baseline

# Volve: Pressure, Water Injection and Oil Production



*Pressure drops with oil production, and recovers under water injection*

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112

months

88

training months

3

input features

## SOURCE & CLEANING

- Source: Volve field dataset, released publicly by Equinor (Equinor Volve data license) then community-processed
- Missing values: No missing months in the Sept, 2007 – Dec, 2016 window used
- Transforms: Cumulative production and injection volumes converted to monthly rates via `.diff()`
- One known data-quality issue: Simulation-derived field averages, not gauge data; production starts in 2008-02.

# Chronological Time Split

THE SPLIT- LAST 18 MONTHS HELD OUT

**88 months**  
**(2008-03-01 to 2015-06-01)**

**Train**

**18 months**  
**(2015-07-01 to 2016-12-01)**

**Test (never seen)**

*last 18 months held out – This split matches real use because in real life historical data is used to forecast future values which have not yet been seen.*



## Metric

RMSE in pressure units, psia

MAE in pressure units, psia

RMSE penalises the large misses that trigger interventions; MAE shows typical error



## Baselines

Naive: persistence

Classical: random forest



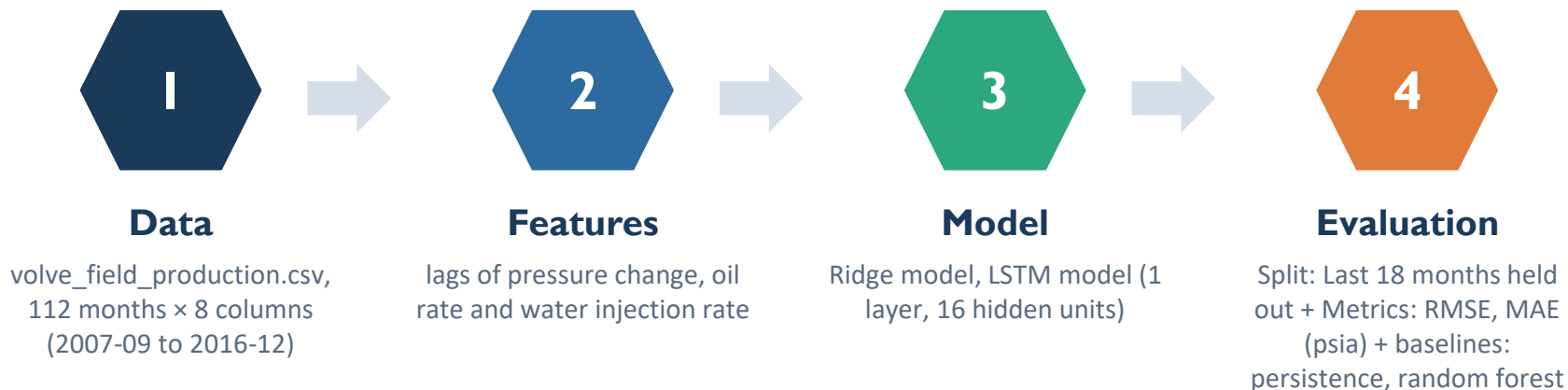
## Leakage checked

Features excluded: GOR, Gas Injection Rate, Gas Production Rate

Scalers fitted on training data only



# Project Pipeline



## TUNING (TRAINING DATA ONLY)

Ridge strength and lag length (3, 6, 12 months) chosen by walk-forward CV (TimeSeriesSplit)  
Scalers fitted on training months only

## WHAT I TRIED THAT DIDN'T WORK

Forecasting oil rate instead of pressure: no model beat persistence  
Adding drivers to the LSTM: no gain on the test months  
6-month recursive forecasts: no better than a flat line

# Results: Best model vs. Baseline

| Model                              | RMSE (psia) | MAE (psia)  | vs. best baseline |
|------------------------------------|-------------|-------------|-------------------|
| Naive baseline (Persistence)       | 85.3        | 49.5        | --                |
| Classical baseline (Random Forest) | 95.1        | 68.4        | -12%              |
| LSTM                               | 90.7        | 72.3        | -6%               |
| <b>Ridge</b>                       | <b>82.0</b> | <b>50.3</b> | <b>+4%</b>        |

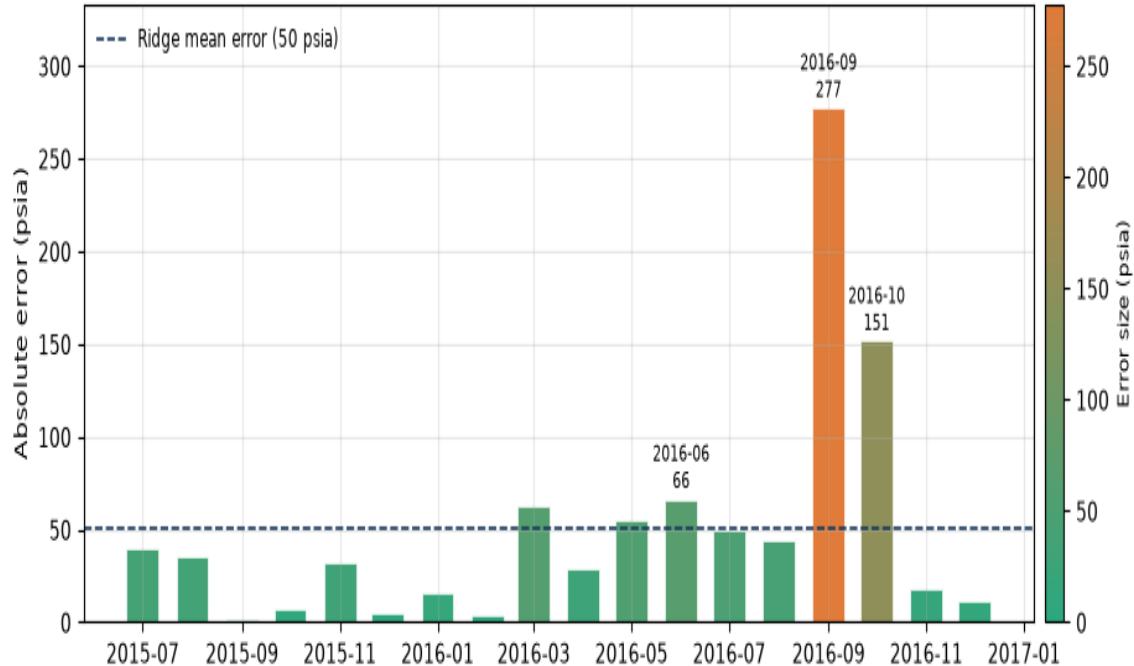
*All rows evaluated on the same split: last 18 months held out. Green row = final model.*

# +4%

3.9 % lower RMSE; MAE not better on unseen test data; Ridge ties with persistence, as slight improvement could be due to noise

# Where it fails: Sudden shut – in

Ridge on dP (LAGS=6) -- pressure + drivers: absolute error by month



## Finding 1

Error was largest at shut in on 2016-09 and 2016-10. It was off by +277 psia and +151 psia



## Finding 2

Inputs show the slowdown but not the shut-in; training data contains no shut-in to learn from.



## Finding 3

Adding a driver that indicates shut in could prove to be useful

# How trustworthy is the Ridge Model?

## HEADLINE UNCERTAINTY

**$\pm 70$  psia**

% prediction interval from walk-forward training errors

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Checked on held-out data: Covers 16 of 18 held-out months (89%); 100% excluding the shut-in

## USE IT FOR

- Month-ahead check of field-average pressure in stable operation, for Volve-type fields
- Benchmarking any fancier forecast against persistence

## DON'T USE IT FOR

- Anticipating shut-ins or rate changes
- Forecasts beyond one month
- Wells or fields other than Volve
- Forecasting oil rate

# Conclusions

- Ridge reaches RMSE 82.0 psia vs 85.3 for persistence on the 18 held-out months.
- The 4% gain is within noise: a tie, not a win.
- The 2016 shut-in causes most of the error, and no model anticipates it.

# Limitations and next steps



## Limitations

- 112 monthly points (88 for training, 18 for testing).
- Two shut-in months (2016-09, 2016-10) produce most of the test RMSE.
- Field-level data derived from a reservoir-simulation history match, not raw gauge data.



## Next steps

- Add an operational flag (planned shut-in / rate-cut schedule)
- Evaluate with rolling-origin backtests over the whole history, not one 18-month window.
- Test well-level pressure data (Volve gauges) rather than a field average.

## THE TAKEAWAY

**The ridge model can forecast pressure after a month to within  $\pm 70$  psia, 90% of the time; but not reliably better than the persistence baseline**

[github.com/sadaraka-kadi/ml4pe-pressure-forecasting](https://github.com/sadaraka-kadi/ml4pe-pressure-forecasting)





# Thank you -- questions?

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# Backup: model card

|                                 |                                                                                                  |                                |                                                                                                                                                           |
|---------------------------------|--------------------------------------------------------------------------------------------------|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Model name &amp; version</b> | volve-pressure-ridge-dP v1.0                                                                     | <b>Evaluation split</b>        | Chronological: last 18 months (2015-07 to 2016-12) held out                                                                                               |
| <b>Intended use</b>             | Month-ahead sanity check on Volve-type field-average pressure for reservoir/production engineers | <b>Headline performance</b>    | RMSE 82.0 psia, MAE 50.3 psia vs persistence 85.3 / 49.5 psia                                                                                             |
| <b>Not intended for</b>         | Forecasting shut-ins or rate changes; horizons beyond 1 month                                    | <b>Where it fails</b>          | 2016-09 and 2016-10 shut-in: errors +277 and +151 psia                                                                                                    |
| <b>Training data</b>            | 88 training months 2008-03 to 2015-06                                                            | <b>Uncertainty</b>             | Empirical 90% interval +/-70 psia                                                                                                                         |
| <b>Inputs &amp; units</b>       | Monthly pressure changes (psia); oil rate production rate and water injection rate (MMSTB/month) | <b>Limitations &amp; risks</b> | Small sample; simulation-derived data; one test window; interval assumes no regime change                                                                 |
| <b>Output &amp; units</b>       | Next-month pressure (psia)                                                                       | <b>How to reproduce</b>        | <a href="https://github.com/sadaraka-kadi/ml4pe-pressure-forecasting">https://github.com/sadaraka-kadi/ml4pe-pressure-forecasting</a> ; Restart & Run All |